Kyra Shi

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SKILLS

Programming Languages: C, C++, Python, Java, Verilog
Embedded Systems: STM32, AVR, FreeRTOS, bare-metal programming, Raspberry Pi
Interfaces & Peripherals: GPIO, I2C, SPI, UART, ADC, PWM, timers, interrupts, BLE
Tools & Debugging: Linux, Git, GDB, SWD, ST-Link, oscilloscope, logic analyzer, DMM

EDUCATION

University of Southern California, Viterbi School of Engineering – Los Angeles, CA **08/2023 - 05/2027**
B.S. Computer Engineering and Computer Science (3.81 GPA)

- Relevant coursework: Embedded Systems, Computer Systems Organization, Operating Systems, Digital Circuits, Smart Devices, Distributed Systems and IoT, Internetworking, Complex Digital ASIC Design
- Involvement: USC Scope Club, USC Presidential Scholar, Viterbi Dean's List

PROJECTS

AVR Embedded Speedometer (github.com/kshisc/EE109_Project**)**

- Developed embedded C firmware to measure object velocity using phototransistors and hardware timers
- Configured AVR timer registers for CTC and Fast PWM generation for sensor, buzzer and servo control
- Implemented a UART driver with register-level configuration and interrupt-driven reception
- Built a 4-bit LCD driver with low-level command/data handling and GPIO control

STM32 Flight Recorder (github.com/kshisc/STM32_Project)

- Designed a 5-task FreeRTOS architecture for sensor sampling, safety monitoring, task supervision, and UI
- Implemented complementary filtering for roll/pitch estimation and tilt/impact detection using an I2C IMU
- Used queue-based logging and mutex-protected data access for thread-safe inter-task communication
- Added heartbeat-based fault detection and conditional hardware watchdog refresh to detect stalled tasks
- Built a UART command shell for runtime calibration and persistent flash storage
- Debugged firmware using ST-Link, SWV trace output, and GDB

Multilevel Queue Scheduler (github.com/kshisc/CS350_Project)

- Created an MLQ scheduler in the xv6 kernel with priority queues, time slicing, and starvation prevention
- Added a system call in C to collect scheduling statistics and monitor process execution
- Analyzed scheduler behavior under diverse workloads, verifying priority boosting and periodic yielding

IoT Environment Monitor (github.com/kshisc/EE250_Project**)**

- Built a Raspberry Pi IoT system with GrovePi sensors to monitor temperature, humidity, and light
- Streamed sensor data to AWS IoT SiteWise via HTTPS/MQTT for dashboard visualization
- Integrated a Flask web interface with dynamic updates powered by an ML classification model

EXPERIENCE

Embedded Systems Lab Assistant – Los Angeles, CA **08/2024 - Present**
University of Southern California

- Supporting 100+ students in coursework covering timers, interrupts, GPIO, UART, and PWM
- Debugging embedded C and bare-metal AVR systems using oscilloscopes and DMMs

Software Engineer Intern – Fremont, CA **05/2024 - 08/2024**
Ridge Security Technology, Inc.

- Resolved critical issues in both the backend and frontend of the company's CRM system
- Built automation tools to generate dynamic sales-order reports, reducing manual effort for operations team
- Created technical documentation to streamline onboarding and knowledge transfer

Robotics Co-Captain – Santa Rosa, CA **01/2021 - 05/2023**
Sonoma Academy (FRC Robotics)

- Integrated hardware and software to control drivetrain motors, sensors, and PID-based autonomous routines
- Extended existing Java and WPILib libraries, using Git for team collaboration